

Decidability of AI Alignment Project

Minerva University

CS142 - Theory of Computation

Prof. Levitt

April 5th, 2026

Will This AI Behave? Good Luck Deciding That

Suppose you've just trained a shiny new language model. It is acing every benchmark, passing the red-team evaluations, and producing perfectly assembled outputs. But then comes the uncomfortable question: is the system actually doing what you want, or is it just very good at looking like it does? A recent paper by Melo et al. (2025) argues that if you're hoping to answer that question by building some kind of universal alignment checker, you're not just out of luck. You're provably out of luck. But the story doesn't end there...

I. The Inner Alignment Problem: Why a Universal “AI Monitor” Can’t Exist

Before getting into the impossibility result, it helps to separate two problems that are often conflated. Outer alignment asks whether we've specified the right goal. Did we write down an objective function that actually reflects our intentions? Inner alignment operates at a different level entirely: even if we have a perfectly specified goal, does the AI system genuinely pursue it across all possible inputs?

We can formalize this.

- ❖ An AI model M maps input bytes to output bytes
- ❖ We assume someone has already solved outer alignment and handed us a judge function $J_v(i, o)$ that determines whether output o is “aligned”
- ❖ The goal is to build a verifier V that checks whether for all inputs, $M(i)$ satisfies J_v

Melo et al. present two complementary arguments. Consider that at the end of the day, every AI program is an algorithm, and by the Church-Turing Thesis, it can be modeled as a Turing machine, making computability theory applicable to AI alignment.

1. The first argument mirrors Turing's proof of the Halting Problem, or the result that no general algorithm can decide for every other program whether it stops or loops forever. Following the same logic, assume a perfect verifier V exists. We can then build an adversarial model M' that runs V on its own bytes and does the opposite of what V predicts. If V declares M' is aligned, M' produces a misaligned output. If V declares M' misaligned, M' produces an aligned output. In this sense, the verifier would be wrong no matter what, so V cannot exist. The authors also show this formally through a reduction from the halting problem, demonstrating that a solution to one would imply a solution to the other.
2. A derivation from the Halting Problem is Rice's Theorem, which states that any non-trivial semantic property is undecidable. In this case, the alignment property is clearly non-trivial (some programs satisfy the judge, some don't) and semantic (it's about input-output *behavior*).

Thus, these arguments prove that no TM can decide, for an arbitrary AI, whether it always satisfies J_v . So, rather than trying to verify an arbitrary AI after the fact, what if we build AI that is aligned by construction?

II. Realistic Models: The Case for Halting-Constrained AI

Having established that the problem is undecidable for Turing machines, the authors make an

observation: real computers aren't actually Turing machines. A physical computer with M bits of memory can occupy at most 2^M configurations, making it technically a finite automaton. And finite automata always halt. They cannot loop forever without revisiting a state.

So if we constrain AI models to always terminate and produce an output, the judge function J_v can always be applied to the output, and the alignment becomes decidable. Feed-forward neural networks, for instance, perform a fixed sequence of operations, guaranteeing termination. However, with an input size of L , there would be 2^L possible inputs that need to be verified against the judge. Decidable, yes. Tractable? Not remotely.

So, the authors suggest flipping the paradigm. Instead of training models and then verifying alignment, start with base models proven to satisfy the judge, define operations that preserve alignment, and any AI built through a finite application of these operations is guaranteed to be aligned. As a more practical approach, they propose running the judge function at inference time on every output. If the judge approves, pass the output. If not, return a pre-approved default output instead. In this sense, the system is aligned directly by construction. In fact, Constitutional Classifiers already implement something similar, but because their judge function is learned rather than formally specified, alignment still depends on the coverage of training data. It's worth noting that, although it guarantees alignment, when the judge rejects, the default output is safe but may be uninformative or irrelevant, sacrificing some usefulness.

There is, however, a catch. A single LLM forward pass always halts, sure. But modern AI systems embed LLMs in loops. An agent's output feeds back as the next call's input, and there is

no guarantee that this outer loop terminates. The very thing that makes agents powerful (iterative reasoning, tool use, planning) is what breaks the halting constraint.

The proposed solution is a θ parameter that architecturally forces the model toward a terminal state regardless of input. This can be thought of as AI's mortality. A mechanism that ensures the system “ages,” eventually terminating. This can be implemented by using self-referential weights, or utility functions incorporating temporal penalties. Test-Time Training (TTT) architectures, in which internal loops converge to a stable fixed point after sufficient steps, are a concrete approach to satisfying this constraint.

III. Practical Critique: Is This Framework Feasible?

The theoretical contributions of the paper are valuable. They prove undecidability, and the reframing from "verify arbitrary AI" to "build aligned AI from certified components" is a useful conceptual shift. In fact, safety-critical fields like nuclear and avionic engineering already operate on this principle. Melo et al. propose extending that logic to AI, and the principle is sound.

However, there are some practical problems that deserve more attention.

First, even for halting-constrained models, exhaustive verification remains intractable. The paper's own engineering targets of 2^{80} operations are far below the complexity of any useful modern AI system. The authors also acknowledge that “one would need a much larger computer

to ascertain a property from the smaller computer,” but how this scales to systems of practical size remains unanswered. Output filtering sidesteps this in some cases, but it’s not eliminating misaligned behavior, just preventing it from being expressed. It is also shifting the burden toward the reliability and completeness of the judge function. In fact, the entire framework assumes J_v is correctly specified and computable. Yet, as mentioned in the paper, human values may be uncomputable and internally contradictory (e.g. autonomous vehicle dilemma). For narrow, well-defined tasks, specifying J_v is plausible. For general-purpose AI, it might not be.

Moreover, the architectures the authors propose constraining are precisely the ones that make modern AI most capable. Reasoning models deliberately extend computation because doing so improves performance. Agentic systems are among the most commercially valuable AI applications today. The θ parameter that guarantees termination also puts a ceiling on how long the model can think, how much it can explore, and how adaptively it can respond to novel situations. This is a fundamental capacity-safety tradeoff that the paper doesn’t fully resolve. TTT’s are a promising direction, but whether bounded convergence can be achieved without significant capability loss remains an open empirical question.

Overall, Melo et al. definitely deliver a well-argued research direction. For those of us concerned with the long-term behavior of AI systems, the paradigm shift and identifying halting as a necessary precondition are meaningful contributions. Yet, the hard engineering and philosophical work still lies ahead.

Word Count: 1199

Melo, G. A., Máximo, M. R., Soma, N. Y., & Castro, P. A. (2025). Machines that halt resolve the undecidability of artificial intelligence alignment. *Scientific Reports*, 15(1), 15591.